

## # StockPulse – Product Requirements Document

**\*\*Tagline:\*\*** One dashboard for every store and warehouse. Your daily numbers, delivered on WhatsApp.

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### ## 1. Executive Summary

StockPulse is a WhatsApp-native inventory and price-intelligence tool built for small and medium business (SMB) owners in Pakistan who run multiple sales channels – online stores and a physical warehouse/shop – and currently have no single, reliable way to see their stock, track supplier price changes, or know what's happening in their business without manually checking each channel separately.

The product combines a simple multi-channel dashboard, AI-driven purchase-price anomaly detection, and two-way WhatsApp interaction, so an owner never has to open an app to know the state of their business – their business comes to them.

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### ## 2. The Problem

- SMB owners running 2-3 online stores plus a physical warehouse have **\*\*no single view of stock\*\*** – they check each channel separately, manually, every day
- **\*\*Price creep from suppliers goes unnoticed\*\*** until it has already cost real money – there is currently no system that tracks purchase-price changes item by item and flags them
- Owners are **\*\*time-poor and not software-first\*\*** – they don't want to learn a new dashboard or log into an app daily; they already live in WhatsApp for nearly everything else in their business
- Existing inventory software is either **\*\*too expensive\*\*** (global enterprise tools), **\*\*not built for Pakistan\*\*** (India-focused tools), or **\*\*not proactive\*\*** (reports exist, but nobody tells the owner something is wrong until they go looking for it)

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### ## 3. Ideal Customer Persona (ICP)

**\*\*Name (persona):\*\*** Someone – owner, General Store & 2 online storefronts

- **\*\*Age:\*\*** 30-45
- **\*\*Business:\*\*** Runs a physical shop/warehouse plus 2-3 online storefronts  
(could be grocery, clothing, electronics – category varies, structure doesn't)
- **\*\*Team:\*\*** 1-3 staff helping with stock and sales, no dedicated IT or accounts person
- **\*\*Current process:\*\*** Tracks stock across a mix of memory, a notebook,

- maybe one spreadsheet – nothing connected, nothing automatic
- **\*\*Relationship with technology:\*\*** Comfortable with WhatsApp daily, not comfortable with unfamiliar business software or dashboards with a learning curve
- **\*\*Core pain:\*\*** Doesn't find out a supplier raised prices until doing monthly totals – by then the cost is already absorbed. Doesn't know exactly what's in the warehouse versus what's actually on the shop floor without physically checking or calling someone
- **\*\*What motivates a purchase decision:\*\*** Trust and simplicity over features. Will not adopt something that requires training or feels like "enterprise software." Price-sensitive – needs to see the cost clearly justified against what they're currently losing to unnoticed price changes and wasted time
- **\*\*Quote (persona voice):\*\*** **"I don't have time to check four different places to know what's going on. Just tell me on WhatsApp."**

### Why this persona is accurate, not a guess

This persona is built directly from the actual gap identified against existing competitors (Section 9) – global tools (Zoho, Extensiv) are priced and designed for businesses with dedicated staff to run them; India-focused tools (Vyapar, Syncli) assume GST/UPI workflows that don't exist in Pakistan. The persona reflects the specific owner these tools currently fail: WhatsApp-fluent, time-poor, price-sensitive, and running a hybrid online+physical operation rather than a single clean channel. This will be validated directly with 5-10 real shop owners during the pilot (Section 14) before being treated as final.

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## ## 4. The Solution

One system that watches stock and prices across every store and the warehouse, and sends the owner what they need to know on WhatsApp – automatically, in their own language, without requiring them to open an app.

### Why WhatsApp specifically

Because the target customer already lives there. Every competing tool requires the owner to change their behavior (log into a dashboard); StockPulse fits into behavior that already exists. This is the single biggest reason this approach is more accurate for this ICP than a traditional dashboard-first product.

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## ## 5. Core Features (Complete Detail)

### 5.1 Multi-Channel Dashboard

- Single view pulling stock and sales data from every connected online store plus the physical warehouse/shop
- Built for non-technical users: every screen leads with a plain-language status line (e.g. "Everything looks fine today") before any raw numbers – see Section 8 for full UX rationale
- Key stats always visible: total stock value, total distinct/different

- items, active price alerts, low-stock count
- **\*\*Item breakdown view:\*\*** clicking into "different items" opens a full list showing, per product, the total quantity and its exact split between warehouse and shop floor (e.g. "Biscuits – 48 total: Warehouse 30, Shop 18") – so the owner always knows not just how much stock exists, but *where* it physically is
- **\*\*Low stock view:\*\*** clicking into "low stock" opens a filtered list of exactly which items have fallen below their restock threshold

### ### 5.2 WhatsApp Integration – Daily Report

- Automatically generated and sent once per day (evening, configurable)
- Contains: stock snapshot, biggest price change of the day, one actionable flag (e.g. "reorder chips – 3 days of stock left at current pace")
- Sent in the owner's chosen language (see 5.7)
- Delivered as a Utility-category WhatsApp message (proactive, business-initiated)

### ### 5.3 WhatsApp Integration – Two-Way Q&A

- Owner can message the number directly – e.g. "biscuits ka stock kitna hai?" or "is mahine sabse zyada profit kis product se hua?" – and receive an instant, accurate answer pulled from their real data
- This is what makes the product feel like "someone watching the shop," not just a report generator
- Classified as a Service-category message under WhatsApp's messaging rules (see Section 11 for cost implications)

### ### 5.4 AI Price-Anomaly Detection

- The moment a new purchase price is entered for any product, it is automatically compared against the last several purchases of that same item
- If the price has moved beyond a set threshold (e.g.  $\pm 10\%$ ), an alert is triggered **\*\*immediately\*\*** – not surfaced only in a monthly report – catching supplier overcharging, data-entry mistakes, or genuine market price shifts while they're still actionable
- This is proactive by design: the owner is told, they don't have to go looking

### ### 5.5 Price Comparison – This Month vs Last Month

- Every product has a price history, comparing current purchase price against previous months
- **\*\*In the daily/weekly WhatsApp message:\*\*** a short text line with a directional indicator – e.g. "📈 Biscuits: PKR 145, up 12% from last purchase" – fast to read, no image required
- **\*\*Weekly (not daily):\*\*** a generated chart image (bar comparison across recent months, current month highlighted) sent as a WhatsApp attachment, giving a fuller visual picture without overloading the owner with a chart every single day
- **\*\*In the dashboard:\*\*** a dedicated Price History screen per product, showing the same bar comparison plus explicit stat rows for current price and % change

### ### 5.6 Data Backup

- Automatic daily backup of all stock, price, and account data, stored

- separately from the live database
- Backup status is surfaced directly and visibly (not buried) – the dashboard and Settings screen both show "Last backed up: [date/time]" with a visible confirmation checkmark, so the owner has proof, not just a promise
- Purpose: protect against data loss, corruption, or server issues – the owner should never be able to lose their business records

### ### 5.7 Language Selection

- Chosen once during first-time setup, changeable anytime afterward in Settings
- Full dropdown of supported languages: English, Urdu, Chinese, German, Arabic, and others – not a limited hardcoded set
- The selected language applies everywhere the owner interacts with the product: WhatsApp daily reports, price alerts, Q&A replies, and the dashboard interface itself
- Dropdown interaction: collapsed by default (e.g. "English ▼"), opens into a clean list with a smooth, soft hover highlight – this is a deliberate UX detail, not incidental

### ### 5.8 Multi-Tenant Architecture (Data Isolation)

- One shared core application ("skeleton") – same features, same code, same structure – used by every shop owner
- Every shop owner's data is completely private and isolated from every other owner's data. A grocery owner's biscuits/juices/chocolates and a clothing owner's shirts/sizes never mix, never become visible to each other, and are stored in entirely separate data spaces
- Each owner defines their own product categories inside their own private space – the system does not assume or hardcode a business type, so it works identically well for groceries, clothing, or any other retail category
- This is standard SaaS multi-tenancy: one product, many private users

### ### 5.9 Staff / Role-Based Access

- Owner can add staff by their WhatsApp number directly – no separate account creation or password system
- Each staff member is assigned a permission level (e.g. "Can add stock" vs. "View only")
- Access list is visible in a dedicated Staff Access screen, showing who was added, when, and their current permission level

### ### 5.10 AI Integration – Where AI Is Actually Used

- **\*\*Price-anomaly detection\*\*** (5.4): AI evaluates each new purchase price against historical purchase data for that item and flags meaningful deviations
- **\*\*Natural-language WhatsApp Q&A\*\*** (5.3): AI interprets the owner's plain-language question (in their selected language) and returns an accurate answer sourced from their real, private data – not a generic/hallucinated response
- **\*\*Daily report generation\*\*** (5.2): AI drafts the plain-language summary and selects the single most relevant actionable flag for that day, rather than dumping every data point
- AI is used to reduce what the owner has to read and interpret, not to add complexity – every AI-generated output is short, direct, and

actionable

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## ## 6. Screens (Confirmed Set)

| Screen         | Purpose                                                                                                                  |
|----------------|--------------------------------------------------------------------------------------------------------------------------|
| Setup          | First-run: language, business name, getting-started checklist, sample-data trial option                                  |
| Dashboard      | Main screen – status line, hero stock value stat, different items / price alerts / low stock tiles, recent price changes |
| Item Breakdown | Full item list with warehouse vs. shop quantity split                                                                    |
| Low Stock      | Filtered list of items below restock threshold                                                                           |
| Add Product    | Product name, category (free text), location, quantity, purchase price, optional variant                                 |
| Price History  | Per-product chart, this month vs. last month, % change                                                                   |
| Staff Access   | Add/view staff by WhatsApp number and permission level                                                                   |
| Settings       | Language, business name, backup status                                                                                   |

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## ## 7. Why Customers Want This (Value Proposition, Explicit)

1. **It removes a daily chore.** Owners currently check multiple channels manually – this replaces that with one automatic message
2. **It catches losses they can't currently see.** Price creep is invisible until a monthly total; this catches it the same day it happens
3. **It requires no new habit.** No new app to remember to open – it uses WhatsApp, which the owner already checks constantly
4. **It's built for their actual language and business context** – not a global or India-first tool translated after the fact
5. **It's affordable relative to what it replaces** – cheaper than enterprise tools, while doing the one thing that actually matters to this owner (visibility + proactive alerts), rather than a long feature list they'll never use

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## ## 8. Competitive Landscape

| Product                | Strength                       | Why it falls short for this ICP                                         |
|------------------------|--------------------------------|-------------------------------------------------------------------------|
| Zoho Inventory         | Mature multi-channel dashboard | Enterprise-priced (PKR 5,600+/month), English-only, not WhatsApp-native |
| Extensiv / Brightpearl | Strong cross-channel reporting | Same – global/enterprise pricing and complexity                         |
| Vyapar                 | WhatsApp-integrated, invoicing | Built around India's GST/UPI ecosystem, not Pakistan's                  |
| Syncli                 | WhatsApp catalog sync          | Same India-first limitation                                             |

**\*\*The gap:\*\*** no existing product combines WhatsApp-first two-way interaction, proactive AI price alerts, and Pakistan-specific setup (language, currency, payment methods) in one product. This is the specific, validated opening StockPulse is built for.

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## ## 9. WhatsApp BSP Strategy

- **\*\*Pilot phase:\*\*** Twilio – free sandbox testing, pay-as-you-go, no upfront cost, fits a developer-led build
- **\*\*Post-pilot, going live in Pakistan:\*\*** switch to WeTarseel – PKR-native billing, local support, ~PKR 14,000/month platform fee

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## ## 10. Non-Functional Requirements

- **\*\*Data isolation:\*\*** strict multi-tenant separation (Section 5.8) – no shop owner's data is ever visible to another
- **\*\*Reliability:\*\*** automatic daily backups (Section 5.6), no data loss even in the event of a crash or corruption
- **\*\*Accessibility for non-technical users:\*\*** every screen must be usable by someone with no prior business-software experience – plain-language status lines, no jargon, no unexplained icons
- **\*\*Performance:\*\*** WhatsApp responses (Q&A) should feel near-instant; the dashboard should load quickly even on modest mobile connections, since owners may check it on their phone
- **\*\*Responsiveness:\*\*** fully usable on mobile, not just desktop

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## ## 11. Demo / Pilot Plan

**\*\*Goal:\*\*** validate the two core hypotheses – that owners actually want proactive price alerts, and that WhatsApp-native interaction beats another login – before building the full product.

**\*\*In scope for the pilot:\*\***

- Manual stock/price entry for one store/warehouse per owner
- Daily WhatsApp report
- Price-anomaly alert on new price entry
- Basic WhatsApp Q&A
- Simple database (not a large production system)
- Twilio free sandbox for WhatsApp testing (zero pilot cost)

**\*\*Out of scope for the pilot (later phases):\*\***

- Real multi-channel API integrations (live store syncing)
- Voice-note stock entry
- Supplier comparison
- Shrinkage/mismatch detection

**\*\*Who:\*\*** 5-10 real shop owners, recruited directly, given hands-on access

and asked for honest feedback – pricing may be offered free during this phase specifically in exchange for candid input

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## ## 12. Success Metrics (Pilot)

- % of pilot owners who check the WhatsApp report without being reminded
- Number of unprompted Q&A questions asked per owner per week (signals real engagement, not just passive reading)
- Number of genuine price anomalies caught that the owner confirms they would not have otherwise noticed
- Willingness-to-pay signal from direct conversation at the end of the pilot period, against the proposed pricing tiers

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## ## 13. Risks & Open Items

- **\*\*Pricing is not yet validated\*\*** with real owners – treated as a hypothesis until the pilot confirms it
- **\*\*WhatsApp Service message pricing change (Oct 1, 2026)\*\*** affects live cost structure – needs to be re-checked against actual BSP billing once past the pilot
- **\*\*Customer acquisition\*\*** – which 5-10 shop owners, and how they're approached, is not yet finalized
- **\*\*Scaling data isolation and backup infrastructure\*\*** beyond pilot scale has not yet been architected in detail – current plan is sufficient for 5-10 pilot users, will need revisiting before wider launch

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\*This document reflects every decision made through the current planning stage. It should be treated as the source of truth for scope – any new decision should be added here, not left undocumented.\*